Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/07/2018
PI Name:	Asadi, Ebrahim Ph.D.		
Project Title:	Enhancement and Verification of Quantitative Phase-Field Crystal Modeling using NASA-PSI Coarsening in Solid- Liquid Mixtures Experiments Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	easadi@memphis.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	901-678-3332
Organization Name:	University of Memphis		
PI Address 1:	Mechanical Engineering		
PI Address 2:	Es310A		
PI Web Page:			
City:	Memphis	**State:**	TN
Zip Code:	38152-0001	**Congressional District:**	9
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	01/01/2017	**End Date:**	12/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
Contact Email:	jan.r.rogers@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Change in period of performance to 1/01/2017-12/31/2018 (originally 9/19/2016-9/18/2018) per PI and NSSC information (Ed., 9/20/17)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Laradji, Mohamed Ph.D. (University of Memphis)		
Grant/Contract No.:	NNX16AT77G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Verification and enhancement of predictive computational material models for solid-liquid mixtures based on experiments conducted on Earth is challenging because of the detrimental gravitational effects on solid particles and liquid mixtures, e.g., buoyancy-induced convection and particle sedimentation. Microgravity environment for NASA-PSI (Physical Sciences Informatics) “coarsening in solid-liquid mixtures experiments” provides data without the detrimental gravitational effects that can be used for verification and enhancement of predictive models. In return, the verified/enhanced model can be used to predict coarsening in solid-liquid mixtures that ideally eliminates the need to conduct similar experiments for other materials and also experiments for other phenomena involved with solid-liquid coexisting. The mentioned predictive capability is only possible if the computational model is not derived phenomenologically for the specific application and material; phase-field crystals (PFC) are a class of such predictive computational models. Therefore, the objectives of this proposal are: 1) The development of a quantitative multi-time and length-scale computational modeling framework and tool around PFC to predict coarsening in Pb-Sn solid-liquid mixtures. Challenges for this objective are quantifying the PFC model and developing an efficient computational tool for three-dimensional (3D) simulations. 2) Data mining of NASA-PSI Coarsening in Solid-Liquid Mixtures (CSLM), CSLM-2, CSLM-2R, and CSLM-3 experiments and using them in series of simulations in the developed computational tool for enhancement and verification of the proposed modeling framework. The challenge for this objective is the huge number of 3D simulations that require significant amount of computer power. 3) Quantifying the developed model and tool to Al-Cu solid-liquid mixtures to show the potential benefits of the proposed research. The challenge for this objective is only the quantification process of the model.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed computational tool verified/quantified based on CSLM data will establish a virtual laboratory that can be used to study coarsening as well as other material processing phenomena on Earth involving with solid-liquid coexistence; e.g,. rapid solidification and grain growth. Since computational models are highly repeatable at lower cost, the proposed modeling framework will result in accelerated material process development and certification. Since solid-liquid coexistence is the initial step in many manufacturing technologies such as casting, liquid infiltration, and additive manufacturing, this modeling framework can be used to investigate the relations between these manufacturing processes and the material properties in order to optimize/control the manufacturing processes parameters. In addition, the results of this research will be disseminated to the public in forms of peer-reviewed articles, scientific reports, a project-specific website, and data uploaded to NASA-PSI website.
Task Progress:	Accomplishments to date are as follows: 1) A literature review on available modified embedded-atom method (MEAM) interatomic potentials for Pb, Sn, Al, and Cu has been completed. Findings: There are MEAM potentials for these pure elements; Pb and Sn MEAM potentials need significant improvements to enhance their accuracy for solid-liquid simulations; and Cu MEAM potential needs minor improvements. There are no binary MEAM potentials for Pb-Sn nor Al-Cu alloy system developed for solid-liquid simulations. 2) MEAM interatomic potentials for pure Pb, Sn, Al, and Cu have been developed. Results: These potentials are capable of predicting solid-liquid coexisting properties such as melting point and solid-liquid interface free energies and high temperature thermoelastic properties in reasonable agreement with their experimental counterparts. 3) A framework to develop MEAM potential for binary alloy systems (Pb-Sn, and Al-Cu) has been established. Developments: This framework integrates free energy calculations at high temperatures with MD simulations to calculate the phase diagram of the system. Since potential parameters for pure elements are already developed (13 parameters), we will next need to determine the coupling parameters using this framework (5 parameters for a system). 4) One-mode PFC model was quantified for pure Pb, Al, and Cu. Outcomes: The effort to quantify the one-mode PFC model for Sn was not successful due to its complex tetragonal crystal structures. As a result, a two-mode PFC model was developed that shows stable Sn crystal structure. This two-mode PFC model will be used for the rest of the project. 5) Binary PFC model (based on the extension of one-mode PFC model for unary materials) was quantified to achieve Pb-Sn phase diagram. Outcomes: Our quantification process was successful to obtain experimental eutectic point and the liquidus curve. However, it was not possible to achieve the solidus curve nor the Sn crystal structure. We expect that using the two-mode binary PFC model with some modifications will overcome these problems. 6) A two-dimensional numerical algorithm using MPI (the message passing interface) protocols was developed to solve unary PFC equations. Developments: This numerical algorithm shows near one-to-one increment of performance by increment of number of processors. We are currently testing the performance of the numerical algorithm over multiple nodes.
Bibliography Type:	Description: (Last Updated: 07/29/2025)
Abstracts for Journals and Proceedings	Nourian-Avval A, Etesami SA, Moats K, Laradji M, Asadi E. "Quantitative Phase-Field Crystal Model for Coarsening in Pb-Sn Solid-Liquid Mixtures." TMS 2017. 146th Annual Meeting, The Minerals, Metals and Materials Society, San Diego, CA, February 26-March 2, 2017. TMS 2017. 146th Annual Meeting, The Minerals, Metals and Materials Society, San Diego, CA, February 26-March 2, 2017. http://www.programmaster.org/PM/PM.nsf/ApprovedAbstracts/DA5CBDB19CB3FF0785257FE3005B34D6?OpenDocument ; accessed 5/7/18. , Feb-2017
Abstracts for Journals and Proceedings	Etesami SA, Asadi E. "Thermomechanical Properties of Metals during Solidification by Molecular Dynamics Simulations." TMS 2017. 146th Annual Meeting, The Minerals, Metals and Materials Society, San Diego, CA, February 26-March 2, 2017. TMS 2017. 146th Annual Meeting, The Minerals, Metals and Materials Society, San Diego, CA, February 26-March 2, 2017. http://www.programmaster.org/PM/PM.nsf/ApprovedAbstracts/70E2F54CE4B3B9A885257FE3005AD784?OpenDocument ; accessed 5/7/18. , Feb-2017

Abstracts for Journals and Proceedings	Nourian-Avval A, Asadi E. "Quantitative simulation of Solid-Liquid Coexistence of Tetragonal and HCP Metals using Phase-Field Crystal Model." Materials Science and Technology 2017, Pittsburgh, PA, October 8-12, 2017. Materials Science and Technology 2017, Pittsburgh, PA, October 8-12, 2017. http://www.programmaster.org/PM/PM.nsf/ApprovedAbstracts/F731C197A29223A0852580EC00537B64?OpenDocument ; accessed 5/7/18. , Oct-2017
Abstracts for Journals and Proceedings	Asadi E. "Solid-Liquid Interface in Metals: A Note on Multi-Scale Bridging for Microscale Modeling." American Society of Mechanical Engineers International Mechanical Engineering Congress and Exposition (IMECE), Tampa, FL, November 3-9 2017. American Society of Mechanical Engineers International Mechanical Engineering Congress and Exposition (IMECE), Tampa, FL, November 3-9 2017. , Nov-2017
Articles in Peer-reviewed Journals	Etesami SA, Asadi E. "Molecular dynamics for near melting temperatures simulations of metals using modified embedded-atom method." Journal of Physics and Chemistry of Solids. 2018 Jan;112:61-72. https://doi.org/10.1016/j.jpcs.2017.09.001 , Jan-2018
Articles in Peer-reviewed Journals	Nourian-Avval A, Asadi E. "Thermodynamics of FCC metals at melting point in one-mode phase-field crystals model." Computational Materials Science. 2018 Apr 1;145:224-34. https://doi.org/10.1016/j.commatsci.2018.01.012 , Apr-2018